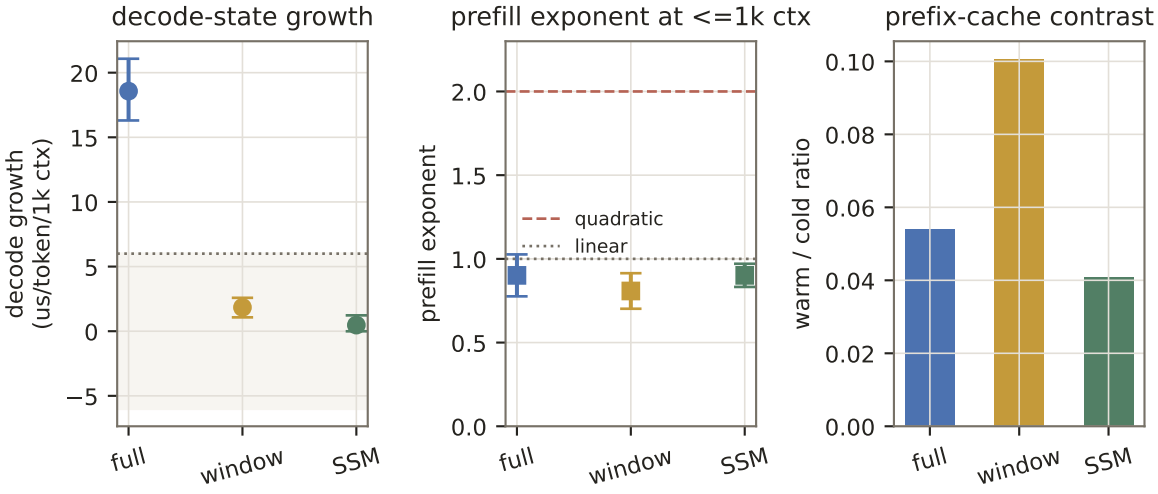
Same-hardware control: what each timing channel can and cannot show
decode-state growth
decode growth
(us/token/1k ctx)
20
15
10
5
0
−5
full
window
SSM
prefill exponent at <=1k ctx
prefill exponent
2.0
1.5
1.0
0.5
0.0
quadratic
linear
full
window
SSM
prefix-cache contrast
warm / cold ratio
0.10
0.08
0.06
0.04
0.02
0.00
full
window
SSM